

GAGANDEEP SINGH

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Software & Data Engineer | Python, Java, ETL Pipelines, Cloud Expertise | Building Scalable, Data-Driven Solutions

About:

I am an experienced Software Developer and Data Engineer with a strong foundation in programming, data pipeline development, and analytics. With expertise in Python, Java, SQL, and modern frameworks like Django and React, I specialize in creating scalable systems that drive efficiency and insights.

Currently pursuing an M.S. in Data Science & Machine Learning (2024), I am skilled in cloud technologies (AWS), database management, and frameworks like TensorFlow, Docker, and Flask. My academic and professional journey has equipped me to tackle complex challenges with innovative solutions.

Education:

M.S. in Data Science & Machine Learning (Expected 2026)
University of San Diego

B.S. in Computer Science
California State University, Sacramento

Core Competencies:

- **Programming & Development:** Python, Java, C++, Django, React, Angular, Flask
- **Data Engineering:** ETL pipeline design, data modeling, database optimization (SQL, NoSQL), cloud infrastructure (AWS)
- **Machine Learning & Analytics:** TensorFlow, Jupyter, R, predictive modeling, data visualization
- **DevOps & Tools:** Docker, GitHub, Jira, React Native, CI/CD pipelines
- **Soft Skills:** Project management, stakeholder communication, cross-functional collaboration

Key Achievements:

- **Optimized Performance:** Reduced client-specific data upload times by **60%-70%** through ETL pipeline enhancements.
- **Revenue Growth:** Designed platform components that contributed to a **50%-60% increase** in gross income for a SaaS product.
- **Scalable Solutions:** Developed and deployed software that significantly enhanced system efficiency and operational productivity.

PROFESSIONAL WORK HISTORY

Data Engineer, Petlove (November 2023 - Current)

- Designed, developed, and maintained scalable ETL pipelines to process and transform large datasets from diverse sources into structured formats for analytics and reporting.
- Built and optimized data architectures, ensuring high availability, fault tolerance, and efficiency for data storage and retrieval systems.
- Collaborated with cross-functional teams to define data strategies, improve data quality, and enhance overall infrastructure.
- Implemented data governance standards, security protocols, and access controls to ensure compliance and data integrity.
- Utilized cloud platforms (e.g., AWS, Azure, GCP) to deploy and manage data pipelines, databases, and analytics tools.
- Developed and maintained documentation for data workflows, schemas, and system processes.
- Conducted performance tuning and troubleshooting for data processes to ensure efficient resource utilization.
- Monitored and maintained data pipeline health, resolving issues to minimize downtime and maintain data consistency.

Software Developer, Stonelake Animal Hospital (January 2018 - Current)

- Managed network infrastructure- Assisted the PMO Manager in assessing project risks and issues, offering solutions as needed.
- Maintained and monitored project plans, schedules, work hours, budgets, and expenditures.
- Tracked project deadlines and proactively escalated risks to the PMO Manager.
- Successfully managed 1-3 projects concurrently, ensuring efficient project progress.
- Orchestrated logistics and hospitality planning for projects, including facility setup, participant access, and catering for key project events.
- Actively organized, attended, and participated in stakeholder meetings, sometimes serving as a facilitator.
- Efficiently documented meeting actions and decisions, ensuring accountability and follow-up.
- Prepared and delivered presentation materials for project-related meetings.
- Troubleshoot network and system issues.
- Conducted routine maintenance.
- Tested and improved software performance.
- Designed the company logo.
- Achieved a 70%+ gross margin increase.

Software Developer, Travel Fix (December 2022 - December 2023)

- Developed and maintained platform components.

- Increased productivity and gross income by 50-60%.
- Created software documentation.
- Undertook project tasks as required, contributing to project success.
- Drafted project communications and training plans, enhancing project team alignment and knowledge sharing.
- Provided essential administrative support as necessary, contributing to the smooth operation of projects.
- Maintained strict adherence to the Company's Quality Management System (QMS) and relevant domestic and global quality regulations.
- Demonstrated a strong understanding of security, privacy, and compliance principles and ensured compliance across the department.
- Played a pivotal role in ensuring departmental compliance with the QMS, regulations, standards, and procedures.
- Actively tackled other work-related duties as assigned, exhibiting flexibility and adaptability.
- Designed and built new products and features.
- Provided IT support and system maintenance.

Freelance Software Developer & Data Analyst, Esports One (May 2022 - June 2023)

- Managed platform development.
- Analyzed and improved existing systems and databases.
- Implemented software enhancements.
- Tailored solutions to client requirements.
- Stayed updated with emerging technologies.
- Analyzed and translated various investment calculations from the mainframe's ALGO to SQL.
- Optimized the dataloader to upload the client-specific translations into the system to increase the work efficiency by 60-70%.

Projects & Contributions

- **Dynamic ETL Pipeline for Real-Time Data:** Designed an end-to-end pipeline for processing large-scale data in real-time using AWS services.
- **Machine Learning Model Deployment:** Built and deployed predictive models for customer segmentation using Flask and TensorFlow.
- **Scalable SaaS Platform:** Developed modular platform components in Python and Java, leading to significant revenue and user growth.

Let's Connect

I am passionate about leveraging software and data engineering to drive innovation and deliver impactful business outcomes. Let's work together to create transformative technology solutions.